Classical vs. Deep Learning Object Detection in X-Band Marine Radar Imagery

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I. Introduction

X-band marine radar produces plan position indicator (PPI) imagery in which maritime objects—vessels, buoys, debris—appear as bright returns against sea clutter and noise, posing a fundamental detection problem for maritime safety and autonomous navigation.

Classical detection approaches are rooted in core image processing techniques. CFAR detectors [1], [2] perform adaptive *segmentation* by estimating local background statistics. Morphological filtering [5] uses *erosion, dilation, opening, and closing* to isolate target structure. Wavelet decomposition [3] provides *multiresolution analysis* to separate targets from clutter. Otsu thresholding [4] offers automatic *segmentation* via histogram analysis, followed by connected component *object recognition*. Recent deep learning methods report strong results: YOLOv11 achieves mAP@0.5 > 0.95 [8], Marine-Faster R-CNN reports 93.65% recall [7], and U-Net [9] and transformer-based [10] architectures have been adapted for radar imagery. However, **no existing study comprehensively compares the full classical pipeline against modern deep learning detectors on a shared marine radar dataset** [6], [12]. This project fills that gap while directly exercising five core CSCE 763 topics: image enhancement, segmentation, morphological processing, multiresolution analysis, and object recognition.

II. Proposed Methods

Classical (7 methods). CA-, GO-, SO-, and OS-CFAR with 2D sliding reference windows, producing binary masks converted to bounding boxes via connected components. **Morphological pipeline**: Otsu binarization → opening (erosion then dilation) → connected component extraction with region property filtering (minimum area, eccentricity). **Wavelet detection** via 2D stationary wavelet transform: target energy concentrates in high-frequency sub-bands, which are thresholded before connected component extraction. **Otsu + region properties** with seed-point detection (SSOTSU [4]) for the small-target problem.

Deep Learning (4 methods). YOLOv11-OBB with SAHI tiled inference [8] on full-resolution PPI scans. **Faster R-CNN** with radar-specific anchor optimization and scale normalization [7]. **U-Net** for per-pixel segmentation, deriving bounding boxes from predicted masks. **RT-DETR** as a transformer-based end-to-end detector without NMS (novel application to marine radar PPI). All 11 methods receive the same preprocessed input and produce bounding boxes for unified evaluation.

III. Evaluation Plan

Dataset. DLR DAAN [8]: 760 X-band PPI frames (3,040 tiles at 640×640, 11,893 instances) with bounding box, OBB, and segmentation annotations. Temporal splitting by recording session prevents data leakage.

Metrics and analysis. COCO metrics (AP@0.5, AP@0.75, AP@[0.50:0.95], AR) stratified by object size; inference time, FLOPs, and parameter count. TIDE framework [11] decomposes errors into classification, localization, duplicate, background, and missed components, further stratified by range, density, and target size. Deep learning models trained with 3 seeds; pairwise bootstrap testing (10,000 resamples, Bonferroni correction).

IV. Timeline, Responsibilities, and Contributions

Nine working weeks (Feb 17–Apr 27; week 4 is spring break). *JC Vaught* handles GPU infrastructure, all 4 deep learning models, inference benchmarking, Pareto analysis, and the deep learning sections. *Ty Dangerfield* handles data preprocessing, all 7 classical method implementations, CFAR parameter tuning, TIDE error analysis, and the classical sections of the final report. Milestones: data ready (wk 2), methods implemented (wk 5), results collected (wk 7), analysis and report complete (wk 9).

Three expected contributions: (1) the first cross-paradigm benchmark for marine radar PPI detection comparing 11 methods under identical conditions; (2) systematic failure analysis revealing condition-dependent failure modes; (3) accuracy vs. latency Pareto analysis for real-time maritime applications.

References

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